Ioan Dragomir

trupples | 1----------3@gmail.com | +40 7-- --- ---

Bd. Paul Teodorescu, Bucharest / Bd. Nicolae Titulescu, Cluj-Napoca

---

Systems engineer with a passion for **embedded software** and **interdisciplinary engineering**. With 15+ years of programming experience, beginning as a hobbyist and active participant in competitive programming, cybersecurity and robotics competitions, and later as a professional, I have developed expertise in both high- and low-level software. In the past couple of years, I have **pivoted to embedded design** and have worked on both digital and mixed-signal projects, with a current focus on **robotics**. **Main skills and interests:**

- Proficiency in C, Python. Competence in C++, MATLAB, Web technologies, and a basic understanding of multiple other programming paradigms. Attention to secure programming practices and producing clean, maintainable code.
- Firmware development with Zephyr RTOS. Robot software development in ROS 2.
- Deep understanding of low-level computer systems: OS internals, various CPU architectures, FPGA design, high- and low-level emulation, low-level optimizations, computer networking.
- Interest in digital and mixed-signal electronics, SDR, DSP, PCB Design, 3D Printing.
- Technical writing, presentation, and teaching skills. Facilitation of effective meetings, knowledge transfer, conflict resolution, integration, and other soft skills.

Work Experience

Analog Devices Romania — Embedded Software Engineer — Jul 2023 – present

Initially in applications engineering and later focused on robotics development, I got to work on a great variety of broad-market projects, my main achievements being:
- Implementing Zephyr firmware, ROS 2 nodes and reference applications for a line of open-source robotics modules, with communication over CANopen
- Designing, implementing and delivering demos and workshops for trade shows and conferences (e.g. Embedded World 2025), and participation in workshops delivered at the Technical University of Cluj-Napoca
- Designing and maintaining production test procedures for ADI boards
- Designing demo hardware in KiCad (educational_tools m2k colorimeter)

Digilent Ro — R&D Intern — Jul 2022 – Sep 2022

After finishing the 2022 edition of the Digilent Digital Design Club, I pursued an internship, during which I collaborated with a diverse team on testing hardware, upgrading and restructuring reference Vivado, Vitis, Petalinux projects running on Digilent development boards.

Bitdefender — Junior Security Researcher — Jul 2021 – Sep 2021

Worked as part of the Active Threat Control team to write malware detection heuristics that run in both user-space and the Windows kernel-space, requiring contributions to the internal ATC framework and monitoring current malware behavior trends.

Competitions

- **DEF CON CTF Finals**: 2021 (8th place), team PTB_WTL
- **Hack-A-Sat Qualifiers**: 2020 (14th place), 2021 (8th place), team OneSmallHackForMan
- **BattleLab Robotica** 2024, 2025 (participation)
- **NSUCRYPTO** International Olympiad in Cryptography: 2020 (10th individual, silver; 8th team, silver), 2021 (3rd individual, bronze; 5th team, silver), 2022 (2nd team, gold), 2023 (21st team, diploma)

Education

Technical University of Cluj-Napoca — Bachelor's Degree Oct 2020 – Jul 2025

Systems Engineering, Automation and Applied Informatics. Thesis subject: "Multi-channel biopotential data acquisition system with ROS 2, libIIO bindings"

Volunteering & Projects

BEST Cluj-Napoca Oct 2021 - Feb 2025

Actively involved in local and international activities of BEST, I have developed both hard- and soft-skills by organising events for students and participating in high-quality training sessions. Relevant positions:

- **Secretary** in the *30th Board of BEST Cluj-Napoca, 2023 – 2024.* Main responsibilities included managing our cooperation with legal entities, the University, and local authorities, as well as driving the organisation's internal working methods.
- **IT coordinator** in the *31st Management, 2024 – 2025.* Supervised all project teams' IT activity, consisting mainly of web development. Designed a training plan and delivered introductory workshops on web development.
- **IT responsible** for *BattleLab Robotica 2023.* Developed a PHP 8 + React scoreboard app and revamped the competiton's livestream. **IT responsible** for *Romanian JAM 2022.* Developed an NFC bracelet payment system, used via a web app based on Flask and React.
- **Academic responsible** for our *BEST Course in Summer 2022* on *autonomous rovers*, in which I designed and built the WiFi-enabled electric rover and software stack. **Workshop organiser** for the 2023 edition on *quadruped robots*, in which I implemented a set of basic quadruped robots with 2DOF legs, and supporting software.

UNBreakable — Mentor Jun 2020 – Oct 2020

Delivered an introductory computer networking and Linux workshop to students preparing for the first edition of the UNBreakable cyber security competition.

Banat IT — CTF organizer Aug 2019 – Jan 2020

Organized the *HackTM CTF Qualifiers* cyber security contest.

- Designed and built reverse engineering and binary exploitation exercises, automating many parts of the workflow using Python,
- Managed an online scoreboard platform for the contest, built on the LEMP stack
- Used docker to securely set up and orchestrate task servers which the competitors had to hack into,

all leading to a flawless qualification round with 747 participating international teams.

Security Espresso — `jmp 0xc0ffee` organizer Jan 2019 – Jun 2019

Organized a weekly cyber security club in affiliation with Security Espresso and Orange FAB. Taught an introductory course in basic reverse engineering, binary exploitation, and network forensics, taking the form of both presentations, hands-on workshops, and CTF participation.